5.1 Introduction to Probability

1. Probability of an Events

$$P(A) = \frac{\text{Number of favorable outcomes (Outcomes in A)}}{\text{Total number of possible outcomes}}$$

*Note: This counting formula works when all outcomes are equally likely

Range of Value: $0 \leq P(A) \leq 1$

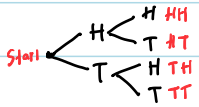
P(A)	Meaning
0	Impossible, it never happens
1	Certain, it always happen (100%)
0.3	30% Chance A happen

2. Sample Space and tree diagram

The sample space S is the set of all possible outcomes.

One coin: $S = \{H, T\}$

Two Coin: $S = \{HH, HT, TH, TT\}$



05.2 Probability Formulas, Symbols & Notations – Marginal, Joint, and Conditional Probabilities

1. Marginal probability – a single event

The probability of one event, regardless of any others:

$$P(A) = \frac{\text{Favorable Outcomes in event } A}{\text{Total Outcomes in Sample Space}}$$

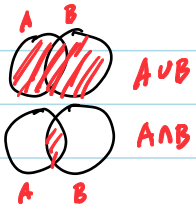
example 1: • Sample Space $S = \{1,2,3,4,5,6,7,8,9\}$

• "Event" $A = \{1,2,3,4\}$ $B = \{6,7,8\}$ $C = \{1,7,8,9\}$

$P(A) = \frac{4}{9} = 44.44\%$ $P(B) = \frac{3}{9} = 33.33\%$. $P(C) = \frac{4}{9} = 44.4\%$

2. Union and Intersection Symbols

Symbols	Name	Meaning
$A \cup B$	Union	Everything in A or B (or both), listed once
$A \cap B$	Intersection	Only what is in both A and B



Example 1:

$A \cup B = \{1,2,3,4,5,6,7,8\}$

$A \cap B = \{\text{None}\}$

$B \cup C = \{1,6,7,8,9\}$

$B \cap C = \{7,8\}$

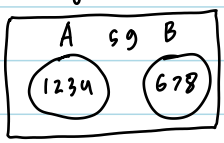
3. Union Probability – "A or B" Addition Rule

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$P(A \cup B)$: $P(A$ or $B)$ — $P(A \cap B)$: $P(A$ and $B)$

The Subtraction stops $A \cap B$ being counted twice.

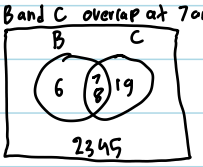
• Mutually exclusive events can't happen together, so $A \cap B \neq \emptyset$ and $P(A \cap B) = 0$: **Mutually Exclusive A, B**



$P(A \cup B) = \frac{7}{9} = 0.778 = P(A) + P(B) = \frac{4}{9} + \frac{3}{9}$

• Example overlap at 7 and 8: $B \cap C = \{7,8\}$ so $P(B \cap C) = \frac{2}{9}$: **Not Mutually Exclusive B, C**

B and C overlap at 7 and 8



$P(B \cup C) = \frac{5}{9} = P(B) + P(C) - P(B \cap C) = \frac{4}{9} + \frac{3}{9} - \frac{2}{9} = \frac{5}{9}$

4. Joint Probability - $P(A \cap B)$ (multiplication rule)

- $P(A \cap B)$ is read "The probability of A and B" (both happen):
  - ↳ Words: $P(A \text{ and } B) = P(A \text{ given } B)\ P(B) = P(B \text{ given } A)\ P(A)$
  - ↳ Symbols: $P(A \cap B) = P(A|B)\,P(B) = P(B|A)\,P(A)$

  On the **Independent events** (one doesn't affect the others): $P(A|B) = P(A)$ and $P(B|A) = P(B)$, so

  Words: $P(A \text{ and } B) = P(A)\,P(B)$

  Symbols: $P(A \cap B) = P(A)\,P(B)$

- Independent or dependent?
  - Situation: Draw a red marble, **Put it back**, then draw blue one

    Type: Independent.

    Why: Replacing the marble restore the bag.
  - Situation: Draw a red marble, **keep it**, then draw blue one

    Type: Dependent.

    Why: The bag now has one fewer marble, so the second probability change.
  - Situation: Toss a coin Twice

    Type: Independent

    Why: The first toss doesn't affect a second toss.

- Notation Warning: $P(A \cap B)$ vs $P(AB)$
  - $P(A \cap B) = P(B \cap A)$: both happen, no order implied.
  - Some books write $P(AB)$ for "A first, then B" and $P(BA)$ for "B first, then A". These are not always equal.
  - Read the context: are the events simultaneous, or does one come before the other?

5. Conditional Probability - "A given B"

$P(A|B)$ is the probability of A **given that** B **has already happened**.

we know that $P(A \cap B) = P(A|B)\,P(B) = P(B|A)\,P(A)$

So

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \qquad P(B|A) = \frac{P(A \cap B)}{P(A)}$$

Conditional = Joint / Marginal

6. Bayes' Theorem

Replacing $P(A \cap B)$ with $P(B|A)\,P(A)$

$$P(A|B) = \frac{P(B|A)\,P(A)}{P(B)}$$

Replacing $P(A \cap B)$ with $P(A|B)\,P(B)$

$$P(B|A) = \frac{P(A|B)\,P(B)}{P(A)}$$

7. Negation (Complement) Probability

Words: P(not A) = 1 - P(A)

Symbols: $P(A') = 1 - P(A)$

$P(A')$ is read "The probability of Not A". The complement is also written A^c (The most common form in textbook), $\bar{A}$ or $\sim A$

example 1:

$P(A') = 1 - P(A) = 1 - \frac{4}{9} = \frac{5}{9}$

$P(B) = 0.35$ then $P(B') = 1 - 0.35 = 0.65$, a 65% chance that B doesnot occur.

>Formula Summary

Type	Words	Symbols
Marginal	$P(A) = \frac{\text{favorable}}{\text{total}}$	(same)
Union (or)	$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
Union, mutually exclusive	$P(A \text{ or } B) = P(A) + P(B)$	$P(A \cup B) = P(A) + P(B)$
Joint (and)	$P(A \text{ and } B) = P(A \text{ given } B)P(B) = P(B \text{ given } A)P(A)$	$P(A \cap B) = P(A \mid B)P(B) = P(B \mid A)P(A)$
Joint, independent	$P(A \text{ and } B) = P(A)P(B)$	$P(A \cap B) = P(A)P(B)$
Conditional	$P(A \text{ given } B) = \frac{P(A \text{ and } B)}{P(B)}$	$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$
Bayes	$P(A \text{ given } B) = \frac{P(B \text{ given } A)P(A)}{P(B)}$	$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}$
Complement	$P(\text{not } A) = 1 - P(A)$	$P(A') = 1 - P(A)$

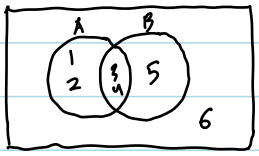
05.3 Probability of Complementary Events

Setup - rolling one six-sided die

- Sample Space $S = \{1,2,3,4,5,6\}$
- A = number at most 4 = $\{1,2,3,4\}$
- $B = \{3,4,5\}$

1. $P(A \text{ and } B)$ - Intersection - $P(A \cap B)$

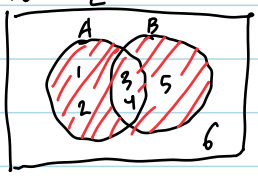
$A \cap B = \{3,4\}$ (the outcomes in both sets)



$$P(A \cap B) = \frac{2}{6} = \frac{1}{3} = 0.33$$

2. $P(A \text{ or } B)$ - Union - $P(A \cup B)$

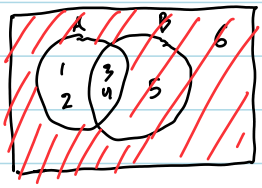
$A \cup B = \{1,2,3,4,5\}$ (the outcomes in either set; only 6 is missing)



$$P(A \cup B) = \frac{5}{6}$$

3. $P(\text{not } A)$ - Complement - $P(A')$

The complement A' is everything in S that is not in A: $A' = \{5,6\}$



4. $P(A)$ and the complement rule

$$P(A) = \frac{4}{6} = \frac{2}{3}$$

$$P(A) + P(A') = \frac{2}{3} + \frac{1}{3} = 1$$

↓ event ↓ complement of event A

An event and its complement together cover the whole sample space, so their probabilities add to 1:

$$P(A') = 1 - P(A)$$

05.4 Conditional Probability with Venn Diagrams and Contigency Tables

- **What it Means**

$P(A|B)$ is the probability that A happens given that B has already happen

Formula: $P(A|B) = \frac{P(A \cap B)}{P(B)}$ Word: $P(A \text{ given } B) = \frac{P(A \text{ and } B)}{P(B)}$

Shortcut: ask "how much of B is also A?" Once B is known to have happened,

$P(A|B) = \frac{\text{Count in both A and B}}{\text{Count in B}}$ → B becomes the new sample space.

- **Example 1 - a die**

$S = \{1,2,3,4,5,6\}$, $A = \text{odd} = \{1,3,5\}$, $B = \{3,4,5\}$

Shortcut: of the 3 outcomes in B, two (3 and 5) are in A:

$$P(A|B) = \frac{2}{3}$$

Formula:

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{2/6}{3/6} = \frac{2}{3} \checkmark$$

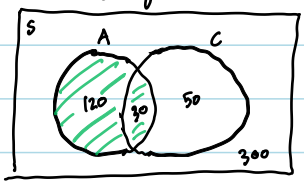
- **Example 2 - Venn diagram (algebra and Chemistry)**

A school has 500 students, 150 take algebra (A), 80 take chemistry (C), and 30 take both.

The four region of the venn diagram (two overlapping circles inside a box of 500):

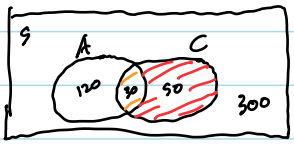
Region	Count
Algebra only (A but not C)	150 − 30 = 120
Both ($A \cap C$, the overlap)	30
Chemistry only (C but not A)	80 − 30 = 50
Neither (outside both circle)	500 − 200 = 300
Total	500

(a) $P(\text{taking Algebra}) = P(A)$



$P(A) = \frac{150}{500} = \frac{3}{10} = 0.3$

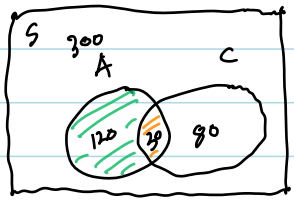
(b) $P(\text{Algebra} \mid \text{Chemistry}) = P(A|C)$



$P(A|C) = \frac{30}{80} = 0.35$

$= \frac{P(A \cap C)}{P(C)} = \frac{30/500}{80/500}$

(c) $P(\text{Chemistry} \mid \text{Algebra}) = P(C|A)$



$P(C|A) = \frac{P(C \cap A)}{P(A)}$

$= \frac{30/500}{150/500} = 0.2$

- Example 3 - Contingency Table (birds in a Zoo)

200 birds: - 70 male with brown eyes $|M \cap Br| = 70$ $n(M \cap Br)$
- 100 femal with brown eyes $|F \cap Br| = 100$ $n(F \cap Br)$
- 20 male with brown eyes $|M \cap Br| = 20$ $n(M \cap Br)$
- 10 female with blue eyes $|F \cap Bl| = 10$ $n(F \cap Br)$

* $|\cdot|$ = Cardinality (number of elements in a set)
* $n(\cdot)$ = Cardinality Function

> Birds

Eye color / Gender	Br Brown Eyes	Bl Blue Eyes	Total
Male	70	20	90
Female	100	10	110
Total	170	30	200

a) P(female)

$P(F) = \frac{110}{200}$

b) P(male and brown eyes)

$P(M \cap Br) = \frac{20}{200}$

c) P(female | brown eyes)

$P(F|Br) = \frac{100}{170}$ or $P(F|Br) = \frac{P(F \cap Br)}{P(Br)} = \frac{100/200}{170/200}$

d) P(blue eyes | Female)

$P(Bl|F) = \frac{10}{110}$

e) P(male | Blue eyes)

$P(M|Bl) = \frac{P(M \cap Bl)}{P(Bl)} = \frac{20/200}{30/200} = 0.67$

Joint vs conditional probability in a contingency table

Orange = numerator. Blue outline = denominator. Light blue = the row or column you restrict to.

(b) P(male and brown)

	Blue	Brown	Total
Male	20	70	**90**
Female	10	100	**110**
Total	**30**	**170**	**200**

70 / 200 = 0.35

joint: divide by the grand total

(c) P(female | brown)

	Blue	Brown	Total
Male	20	70	**90**
Female	10	100	**110**
Total	**30**	**170**	**200**

100 / 170 ≈ 0.588

given brown: divide by the column total

(e) P(blue | female)

	Blue	Brown	Total
Male	20	70	**90**
Female	10	100	**110**
Total	**30**	**170**	**200**

10 / 110 ≈ 0.091

given female: divide by the row total

The numerator is always one cell. What changes is the denominator: the grand total for a joint probability, or one row or column total for a conditional probability.

Key takeaways

- $P(A \mid B) = \dfrac{P(A \cap B)}{P(B)}$, or with counts, $\dfrac{n(A \cap B)}{n(B)}$.
- The "given" event becomes the **denominator**, the new reduced sample space.
- In a contingency table, a conditional probability uses **one row or column total** as the denominator. A joint probability uses the **grand total**.
- $P(A \mid B)$ and $P(B \mid A)$ are usually different.

05.5 Probability of Independent and Dependent Events

1. Notation "A then B" as an Intersection

The video says "P(A then B)". Formally this is an **intersection** ("and"), with the order recorded by subscripts:

A_1 = "A happens on the 1st draw" $\qquad P(A \text{ then } B) = P(A_1 \cap B_2)$

B_2 = "B happens on the 2nd draw"

The symbol $\cap$ means "and". Since $A \cap B = B \cap A$, the order ("then") comes from the subscript, not from $\cap$

- **The Multiplication Rule (General Form)**

From the Definition of Conditional Probability $P(B|A) = \frac{P(A \cap B)}{P(A)}$, multiplying both sides by $P(A)$ gives:

$$P(A_1 \cap B_2) = P(A_1) \times P(B_2|A_1)$$

2. Definitions

- **Independent events:** The outcome of one does **not** change the probability of the other, i.e. $P(B_2|A_1) = P(B_2)$, so the condition drops out:

$$P(A_1 \cap B_2) = P(A_1) \cdot P(B_2)$$

- **Dependent events:** The first outcome **changes** the probability of the second, so the conditional probability must be kept:

$$P(A_1 \cap B_2) = P(A_1) \cdot P(B_2|A_1)$$

Drawing marbles...	Type	Why
with replacement	Independent	The marble goes back, so the bag is unchanged
without replacement	Dependent	The bag has one fewer marble (and maybe one fewer of a color)

- **Example ~ A bag of Marbles**

	Red	Blue	Green	Yellow	Total
n(·)	8	7	6	4	25

* Notation: R_1 = red on draw 1, G_2 = green on draw 2, and so on.

T_2 = The set of marbles available for the second draw.

(a) P(red)

$$P(R) = \frac{8}{25} = 0.32$$

(b) P(Blue, then Green, with replacement)

$$P(B_1 \cap G_2) = \frac{7}{25} \times \frac{6}{25} = \frac{42}{625} = 0.0672$$

(c) P(yellow, then red, without replacement)

$$P(Y_1 \cap R_2) = P(Y_1) \times P(R_2|Y_1) = \frac{n(Y_1)}{n(T_1)} \times \frac{n(R_2)}{n(T_2)} = \frac{4}{25} \times \frac{8}{24} = \frac{4}{75} = 0.0533$$

(d) Two Blue, with replacement

$$P(B_1 \cap B_2) = P(B_1) \cdot P(B_2) = \frac{n(B_1)}{n(T_1)} \times \frac{n(B_2)}{n(T_2)} = \frac{7}{25} \times \frac{7}{25} = 0.0784$$

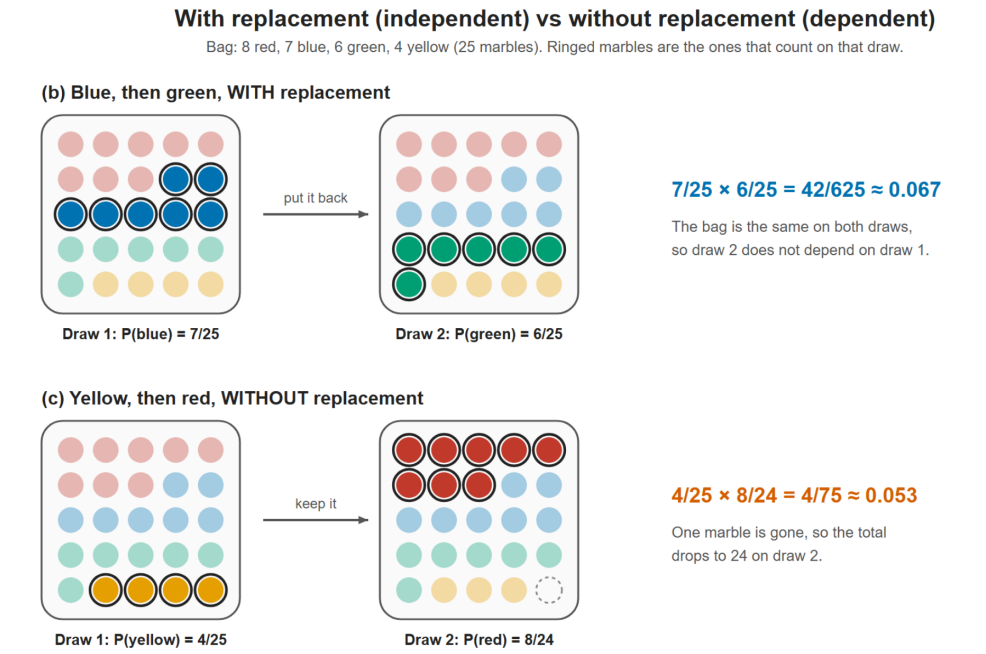
(e) Two Green, without replacement

$$P(G_1 \cap G_2) = P(G_1) \times P(G_2|G_1) = \frac{n(G_1)}{n(T_1)} \times \frac{n(G_2)}{n(T_2)} = \frac{6}{25} \cdot \frac{5}{24} = 0.05$$

After one green marble is removed, there are 5 greens out of 24

Why (b) is independent and (c) is dependent

- In (b), $P(G_2|B_1) = \frac{6}{25}$, which is exactly $P(G_2)$ on its own. The first draw has no effect.
- In (c), $P(R_2|Y_1) = 8/24$, instead of $P(R_2) = \frac{8}{25}$, the first draw changes it.



3. Key Takeaways

- With replacement → Independent: multiply the unchanged probabilities.
- Without replacement → Dependent: after each draw, reduce the total and (if needed) the count of the color drawn.
- "First... then..." problems use the multiplication rule: $P(A_1 \cap B_2) = P(A_1)\, P(B_2|A_1)$
- For three or more draws it chains: $P(A_1 \cap B_2 \cap C_3) = P(A_1) \cdot P(B_2|A_1) \cdot P(C_3|A_1 \cap B_2)$

5.6. Probability of Mutually Exclusive Events with Venn Diagrams

1. Definition

Events A and B are **mutually exclusive** if they **cannot happen at the same time**, meaning they share **no outcomes**:

$A \cap B = \emptyset \iff P(A \text{ and } B) = 0$

Venn Diagram: mutually exclusive events are drawn as **separate, non-overlapping** circles

events that are not mutually exclusive are drawn as **overlapping** circles.

2. Testing for Mutual Exclusivity

One die, $S = \{1,2,3,4,5,6\}$

- $A = \{1,2,3\}$, $B = \{5,6\}$, $C = \{3,4,5\}$

Rule: • If $P(A \cap B) = 0$, They are mutually exclusive.
If it is anything other than 0, They are not.

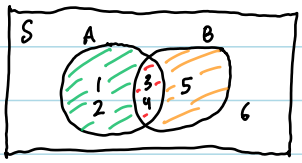
Pair	Intersection	P(both)	Mutually Exclusive?
A, B	$\emptyset$	0	yes
A, C	3	1/6	No
B, C	5	1/6	No

3. The Addition Rule

- **General (always works):** $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- **Mutually Exclusive (The last term is 0):** $P(A \cup B) = P(A) + P(B)$

• Example 1 - Overlapping Events

$A = \{1,2,3,4\}$, $B = \{3,4,5\}$, $S = \{1,2,3,4,5,6\}$

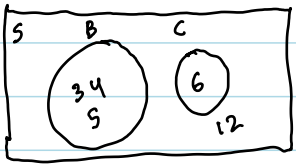


$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$
$$= \frac{4}{6} + \frac{3}{6} - \frac{2}{6} = \frac{5}{6}$$

• Example 2 - Mutually Exclusive Events

$B = \{3,4,5\}$ $C = \{6\}$

$$P(B \cup C) = P(B) + P(C)$$
$$= \frac{3}{6} + \frac{1}{6}$$
$$= 4/6$$



4. Key Takeaways

- Mutually exclusive = no shared outcomes = $P(A \cap B) = 0$
- use $P(A) + P(B) - P(A \cap B)$ in general. For mutually exclusive $P(A \cap B) = 0$ then it will be $P(A) + P(B)$
- A probability above 1 is a sign that an overlap was counted twice.

5.7 Addition Rule and Multiplication Rule

1. Addition Rule - "Or"

$P(A \text{ or } B) = P(A \cup B) = P(A) + P(B) - P(A \cap B)$

- for Mutually Exclusive events $P(A \cap B) = 0 \rightarrow P(A \cup B) = P(A) + P(B)$
- The general form Always works.

2. Multiplication Rule - "And"

Start from Conditional Probability:

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \rightarrow P(A \cap B) = P(B) \times P(A|B)$$

$$P(B|A) = \frac{P(A \cap B)}{P(A)} \rightarrow P(A \cap B) = P(A) \times P(B|A)$$

Independent events: $P(A|B) = P(A)$ and $P(B|A) = P(B)$, so

$P(A \cap B) = P(A) \times P(B)$

Note: "$A \cap B$" means both happen together, so $P(A \cap B) = P(B \cap A)$ this is not the same as "A then B" or "$A_1 \cap B_2$", where Order can Matter.

3. Test

Property	Test
Independent	$P(A\|B) = P(A)$, or equivalently $P(A \cap B) = P(B) \times P(A\|B) = P(A)P(B)$
Mutually Exclusive	$P(A \cap B) = 0$

4. Example - Sarah's Course Choice

- $P(A) = P(\text{enrolls in Algebra}) = 0.3$
- $P(B) = P(\text{enrolls in Biology}) = 0.7$
- $P(A|B) = P(\text{Algebra given Biology}) = 0.4$

* Note: we know that $P(A|B) = \frac{P(A \cap B)}{P(B)}$

(a) $P(\text{algebra and biology})$ - multiplication rule

$P(A \text{ and } B) = P(A \cap B) = P(B) \times P(A|B) = 0.7 \times 0.4 = 0.28$

(b) $P(\text{algebra or biology})$ - addition rule

$P(A \text{ or } B) = P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.3 + 0.7 - 0.28 = 0.72$

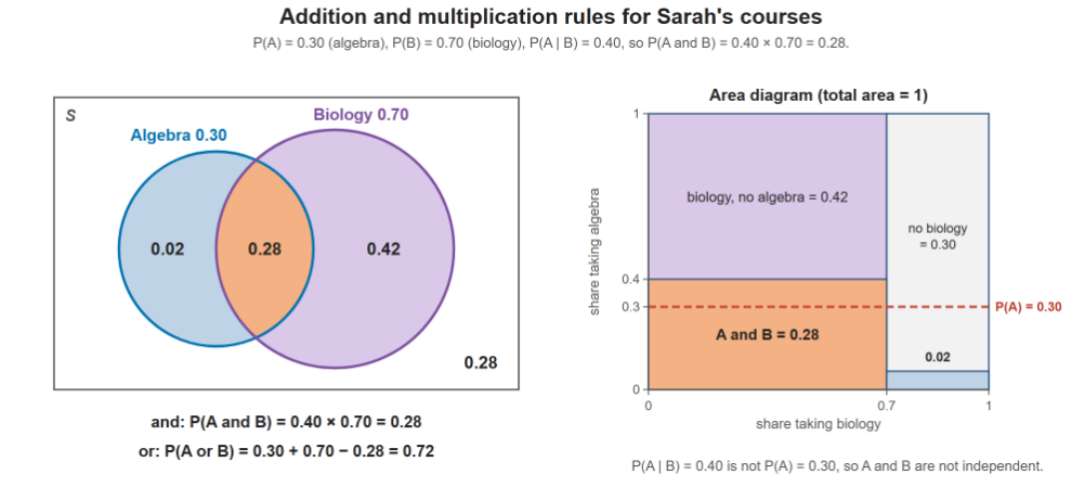
(c) Are the events Independent?

$P(A) \neq P(A|B)$ → so the events are not independent: taking biology makes algebra more likely

(d) Are the events mutually exclusive?

$P(A \cap B) = 0.28 \neq 0$, so they are not mutually

Illustration for Sarah's Courses:



- Key Takeaways
  - Or → add (subtract the overlap). And → multiply (use conditional probability unless the events are independent).
  - Independent ≠ Mutually exclusive:
    - Independent: One doesn't effect the other, $P(A \cap B) = P(A)\,P(B)$
    - Mutually exclusive: They can't both happen, $P(A \cap B) = 0$

5.8 Compound Probability of Independent Events

1. Rule

For **Independent** events, multiply:

$$P(A \cap B) = P(A) \cdot P(B)$$

- **Example 1 Coins**

(a) Two Heads in two flips

$$P(HH) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$

Check with the sample space $\{HH, HT, TH, TT\}$: 1 of 4 outcomes ✓

(b) Heads, tails, heads, in that order, in three flips

$$P(HTH) = \left(\frac{1}{2}\right)^3 = \frac{1}{8}$$

(c) Two tails and one head, in any order

The matching outcomes are HTT, THT, TTH

$$P(2T \text{ and } 1H \text{ in any order}) = \frac{3}{8}$$

Does the order matter? In (b) a specific sequence is required, so only one outcome counts. In (c) any arrangement counts, so add up all the matching outcomes.

- **Example 2 - A standard 52-card deck.**

Suit	Color	Cards
♢ Diamonds	Red	A, 2-10, J, Q, K (13)
♡ Hearts	Red	13
♤ Spades	Black	13
♧ Clubs	Black	13

- 4 Suits × 13 = 52 Cards : 26 Red, 26 Black : No Jokers
- Each rank (e.g Queen) Appear 4 Times, Once per Suit

(a) P(Queen)

$$P(Q) = \frac{4}{52} = 7.69\%$$

(b) P (Black Card Numbers $> 3 \leq 9$)

Number : 4 5 6 7 8 9

$$P = \frac{6 \times 2}{52} = \frac{12}{52} = \frac{3}{13}$$

(c) $P(\text{Red King}_1 \cap \text{Diamond}_2)$ without replacement

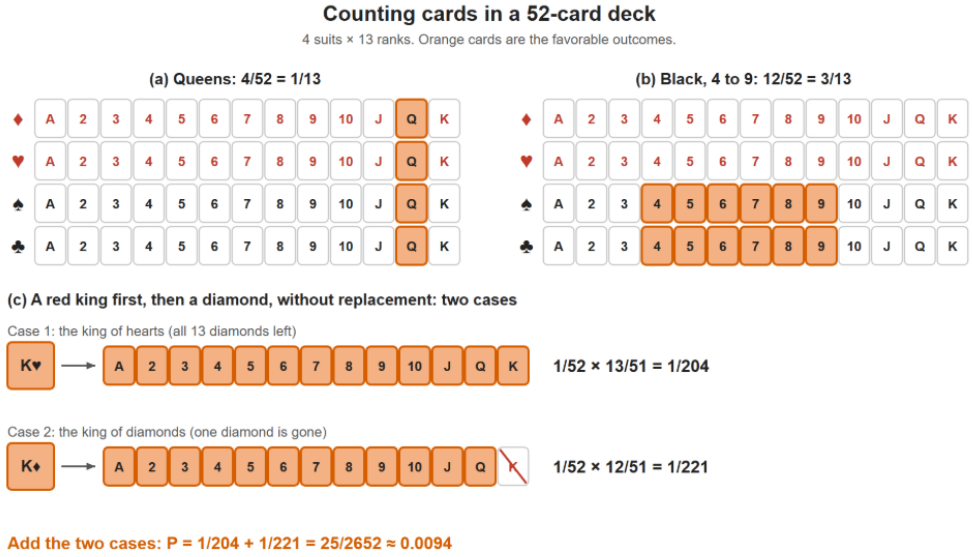
This is **Dependent**: How Many Diamonds are Left Depends on which red king was Drawn. Split it Into Two Cases and Add Them

Case	First Card	Second Card	Probability
1	King ♡ $(1/52)$	any ♢ all 13 are still here $\frac{13}{51}$	$\frac{1}{52} \times \frac{13}{51} = \frac{1}{204}$
2	King ♢ $(1/52)$	any ♢ only 12 are left $\frac{12}{51}$	$\frac{1}{52} \times \frac{12}{51} = \frac{1}{221}$

$$P(KR_1 \cap D_2) = \frac{1}{204} + \frac{1}{221} = 0.0094$$

* Writing $P(KR_1 \cap D_2) = P(RK) \times P(D) = \frac{2}{52} \times \frac{13}{51}$ would be wrong | Because the second probability depends on which king was taken.

Illustration of 52-Card Deck



Counting favorable cases in the deck. For part (c) the number of diamonds left depends on which king was taken first, so the two cases are worked out and separately added.

Key Takeaways

- Independent events: Multiply the probabilities. n fair coin flips in a specific order: $(1/2)^n$
- "In any order" → Count all the arrangements that work.
- Without replacement → Dependent. If the second probability depends on which first outcome happened, split into cases and add.

5.9. How to Calculate Expected Value

1. Formula

The expected value is the long-run average result per trial:

$$E(X) = \sum x\,P(x) = x_1 p_1 + x_2 p_2 + \dots + x_n p_n$$

- X = the value of each outcome (gains positive, losses negative)
- P = its probability, as a Decimal; all the p's add up to 1

Steps: make a table of outcomes → value → probability, multiply each value by its probability and add.

Expected total over n trials = $n \times E(X)$.

2. Example 1 - A Casino Game

Lisa - wins 500 with probability of 20%.

- lose -100 with probability of 80%.

Outcome	Value x	Probability p	x.p
Win	+500	0.2	+100
Lose	-100	0.8	-80
Total		1.00	+20

$$E(X) = 500 \times (0.20) + (-100)(0.80) = 100 - 80 = \$20 \text{ per game}$$

- Over 10 games: $10 \times 20 = \$200$ expected
- Over 100 games: $100 \times 20 = \$2000$ expected

3. Example 2 - Profit per Laptop

Company XYZ-earns 40 on each working laptop it sell and

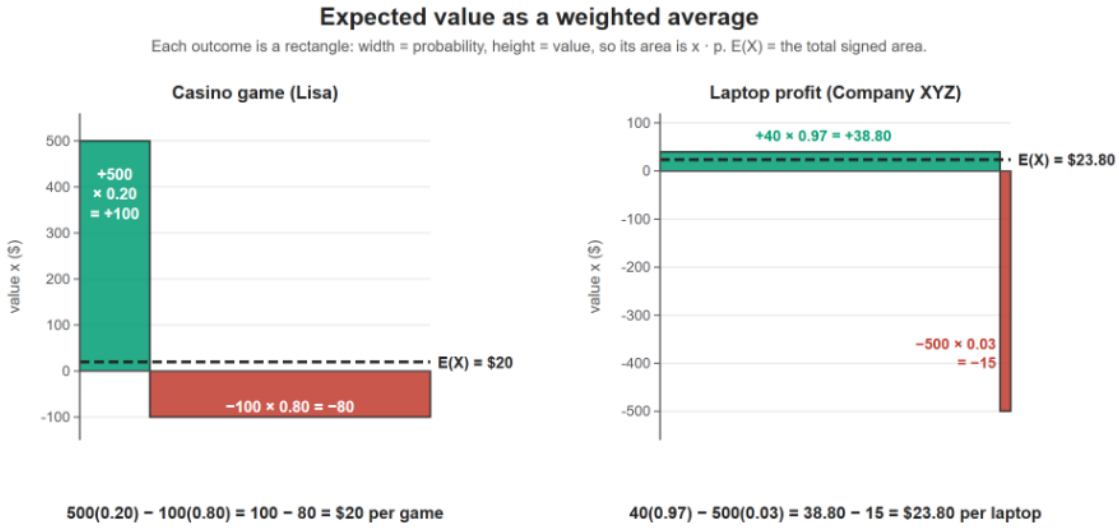
- lose 500 on each working laptop returned as defective. 3 in 100 laptops are defective

Outcome	Value x	Probability p	x.p
Works	+40	0.97	+38.80
Defective	-500	0.03	-15.00
Total		1.00	+23.80

$$E(X) = 40(0.97) + (-500)(0.03) = 38.80 - 15.00 = \$23.80 \text{ per laptop}$$

Selling 100 laptops is expected to earn $100 \times 23.80 = \$2380$

4. Example 1 and 2 Illustration



Each outcome becomes a rectangle with area x times p. Positive areas (green) minus negative areas (red) give E(X). The dashed line at E(X) is the average height over the whole width.

Key Takeaways

- $E(X) = \sum x \cdot P$: weight each outcome's value by its probability.
- Losses go in as negative values
- The expected values is an average over many trials A single game never pays exactly $20
- Total expected = number of trials x expected value per trials

5.10 - Probability Tree Diagrams

1. How a Tree Diagram Works

- Each branch shows one possible outcome of a step, labelled with its probability
- Branches leaving the same point add up to 1
- Along a path ("and") → multiply the probabilities.
- Combining paths ("or") → add the probabilities.
- All the final path probabilities add up to 1.

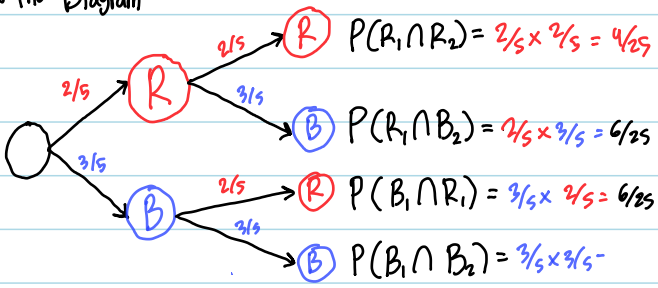
1. Problem 1 - With Replacement (Independent)

A bag holds 8 red and 12 blue marbles (20 total). Two marble are drawn with replacement.

- Calculate the probabilities of each draws

$P(R) = 8/20 = 2/5$, $P(B) = 12/20 = 3/5$

- Draw the Diagram



(a) P(two blue)

$$P(B_1 \cap B_2) = P(B_1) \times P(B_2) = P(B) \times P(B) = 9/25 = 0.36 = 36\%$$

(b) P(at least one Red)

$$P(n(R) \geq 1) = P(R_1 \cap R_2) + P(R_1 \cap B_2) + P(B_1 \cap R_2)$$
$$= \frac{2}{5} \times \frac{2}{5} + \frac{2}{5} \times \frac{3}{5} + \frac{3}{5} \times \frac{2}{5} = \frac{16}{25} = 0.64$$

Complement shortcut: "At least one red" = "not both blue"

$$P(n(R) \geq 1) = P((B_1 \cap B_2)') = 1 - P(B_1 \cap B_2) = 1 - 3/5 \times 3/5 = 16/25 = 0.64$$

(c) $P(R_1 \cap B_2)$

$$P(R_1 \cap B_2) = 2/5 \times 3/5 = 6/25 = 24\%$$

(d) P(one red and one blue)

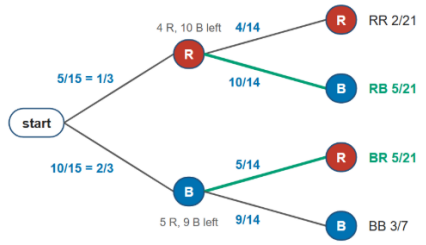
$$P(n(R)=1 \cap n(B)=1) = P(R_1 \cap B_2) + P(B_1 \cap R_2)$$
$$= 2/5 \times 3/5 + 3/5 \times 2/5$$
$$= 12/50 = 0.24$$

2. Problem 2 - Without Replacement (Dependent)

A Bag hols 5 red and 10 blues marble (15 total). Two marbles are draw without Replacement by Sally.
Thus the second-draw probabilities depens on the first draw.

- Calculate the Probabilities of each Draws

$P(R) = 5/15 = 1/3$ $P(B) = 10/15 = 2/3$



5/21 + 5/21 = 10/21 ≈ 0.476
Second-draw branches use 14 marbles and the updated counts.

(a) $P(n(R)=2) = P(R_1 \cap R_2)$

$P(R_1 \cap R_2) = P(R_1) \times P(R_2|R_1) = \frac{5}{15} \times \frac{4}{14} = 0.095$

(b) $P(n(R)=1 \cap n(B)=1)$

$P(n(R)=1 \cap n(B)=1) = P(R_1 \cap B_2) + P(B_1 \cap R_2)$

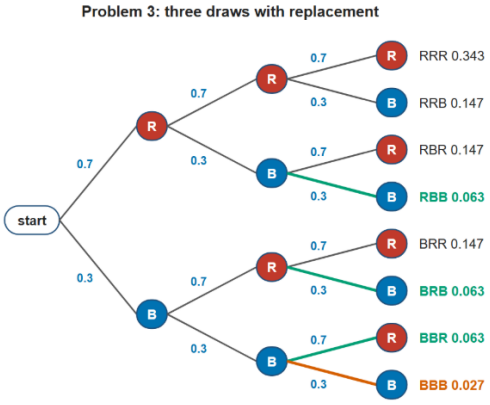
$= P(R_1) \times P(B_2|R_1) + P(B_1) \times P(R_2|B_1)$

$= \frac{5}{15} \times \frac{10}{14} + \frac{10}{15} \times \frac{5}{14} = 0.476$

3. Problem 3 - Three draws with replacement

3 Blue, 7 red (10 total). Megan Draw 3 with replacemen $P(R)=0.7$, $P(B)=0.3$ every time

There are 2^3 = outcome : $R_1 \cap R_2 \cap R_3$, $R_1 \cap R_2 \cap B_3$, $R_1 \cap B_2 \cap B_3$, $B_1 \cap R_2 \cap R_3$, $B_1 \cap R_2 \cap B_3$, $B_1 \cap B_2 \cap R_3$, $B_1 \cap B_2 \cap B_3$.



(c) no red = BBB = $0.3^3 = 0.027$
(d) exactly two blue = 3 × 0.063 = 0.189
(a) at least one red = 1 − 0.027 = 0.973

(a) P(at least one red): every outcome except $B_1 \cap B_2 \cap B_3$

$P(n(R) \geq 1) = 1 - P(B_1 \cap B_2 \cap B_3) = 1 - 0.3^3 = 1 - 0.027 = 97.3\%$

(b) P(at most two blue): 0, 1 or 2 Blue, again everything except $B_1 \cap B_2 \cap B_3$

$P(n(B) \leq 2) = 1 - P(B_1 \cap B_2 \cap B_3) = 97.3\%$

(c) $P(\text{no red}) = P(B_1 \cap B_2 \cap B_3)$

$P(B_1 \cap B_2 \cap B_3) = 0.3^3 = 0.027 = 2.7\%$

(d) $P(n(B) = 2)$

$P(n(B) = 2) = P(B_1 \cap B_2 \cap R_3) + P(B_1 \cap R_2 \cap B_3) + P(R_1 \cap B_2 \cap B_3)$

$= (0.3 \times 0.3 \times 0.7) \times 3 = 0.3 \times 0.3 \times 0.7 \times 3 = 0.189 = 18.9\%$

5.11 Bayes' Theorem of Probability with Tree Diagram

1. Deriving Bayes' Theorem

Start from two Conditional Probabilities $P(A|B) = \frac{P(A \cap B)}{P(B)}$, $P(B|A) = \frac{P(A \cap B)}{P(A)}$

Thus, Both give the same joint probability $P(A \cap B) = P(B) \times P(A|B) = P(A) \times P(B|A)$

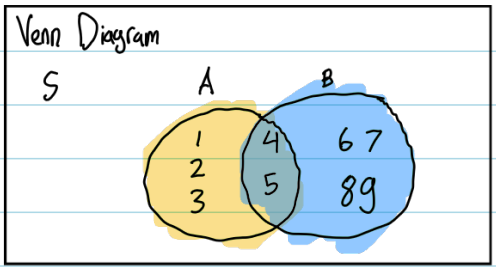
Bayes Theorem is :	$P(A|B) = \frac{P(A) \times P(B|A)}{P(B)}$ (Thus)	$P(B|A) = \frac{P(B) \times P(A|B)}{P(A)}$

2. Example 1 - Slips of Paper (Checking the Formula)

A bottle holds slip numbered 1-9, each equally likely to be drawn.

$A = \{1,2,3,4,5\}$, $B = \{4,5,6,7,8,9\}$, $A \cap B = \{4,5\}$

Venn Diagram



Calculate the Probability:

- $P(A) = 5/9$, $P(B) = 6/9$
- $P(B|A) = \frac{2}{5}$

- Calculate $P(A|B)$ and $P(B|A)$ using bayes Theorem:

$$P(A|B) = \frac{P(A)\,P(B|A)}{P(B)} = \frac{5/9 \times 2/5}{6/9} = 0.333 = \frac{1}{3} \qquad P(B|A) = \frac{P(B)\,P(A|B)}{P(A)} = \frac{6/9 \times 1/3}{5/9} = 0.4 = \frac{2}{5}$$

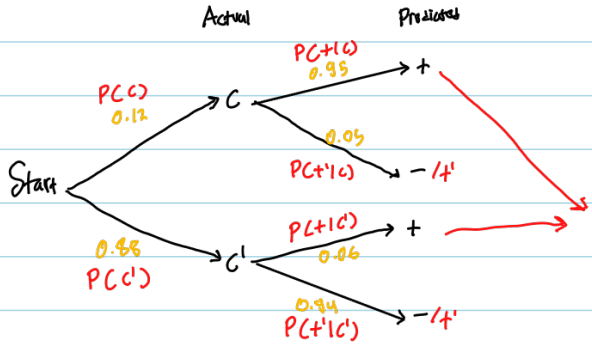
- Check with the direct formula:

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{2/9}{6/9} = \frac{1}{3} \qquad P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{2/9}{5/9} = 0.4 = \frac{2}{5}$$

2. Example 2 - Medical Screening

- 12% of men develop prostate cancer: $P(C) = 0.12$
- With cancer, The test is positive 95% of the time: $P(+|C) = 0.95$
- Without Cancer, There is a 6% false-positive rate : $P(+|C') = 0.06$

The question is: $P(C|+)$ what is the probability of True positive or develop cancer given it testes positive



$P(+'|C) = 1 - P(+|C) = 1 - 0.95 = 0.05$

$P(+'|C') = 1 - P(+|C') = 1 - 0.06 = 0.94$

Thus Probability of Positive

$P(+) = P(C) \times P(+|C) + P(C') \times P(+|C')$

Case C, Case C'

$= 0.12 \times 0.95 + 0.88 \times 0.06$

$P(+) = 0.1668$

$$P(C) \times P(+|C) = P(+) \times P(C|+)$$

$$\text{thus } P(C|+) = \frac{P(C) \times P(+|C)}{P(+)} = \frac{0.12 \times 0.95}{0.1668} = 0.683$$

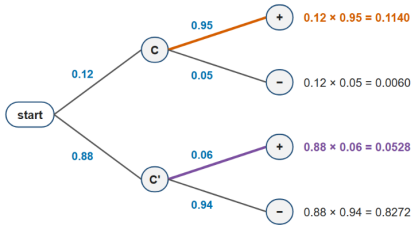
- Same answer with Counts (Imagine 10,000 Men)

	Positive (16.68%)	Negative (83.32%)	Total
Cancer (12%)	1140	60	1200
No Cancer (88%)	528	8272	8800
Total	1668	8332	10000

$$P(C|+) = \frac{1140}{1668} = 0.683 \checkmark$$ → The choice of 10000 don't matter, what matter is the ratio.

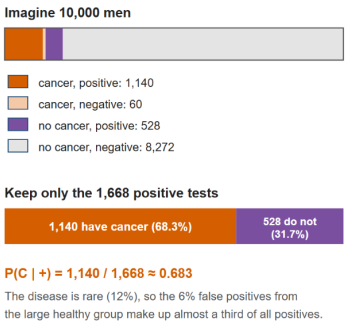
Bayes' theorem: why a positive test is not a sure diagnosis

P(C) = 0.12, P(+ | C) = 0.95, P(+ | no C) = 0.06. Only the positive paths (colored) matter once the test is positive.



P(+) = 0.1140 + 0.0528 = 0.1668

P(C | +) = 0.1140 / 0.1668 ≈ 0.683



P(C | +) = 1,140 / 1,668 ≈ 0.683

The disease is rare (12%), so the 6% false positives from the large healthy group make up almost a third of all positives.

Out of 10,000 men, 1,668 test positive, but only 1,140 of them have cancer. Restricting to the positive tests gives P(C | +) = 1,140 / 1,668 â□□ 0.683.

4. Key Takeaways

- $P(A|B) = \frac{P(A \cap B)}{P(B)}$ which turns a known conditional probability around
- The denominator often has to build from a tree

  $P(B) = P(B|A)P(A) + P(B|A')P(A')$
- A positive test does not guarantee the condition. Rare condition plus false positive can produce many false alarms
- Converting to counts (e.g 10 000 people) is an easy way to check the answer.